THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 9: **Some Applications of Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angle of depression of an object from an observer is 30°. The angle of elevation of the observer as seen from the object is: 1

(a) 0° **(b)** 60° **(c)** 90° **(d)** 30°

2. From a point on level ground, 20 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) $20\sqrt{3}$ m **(b)** $\frac{20\sqrt{3}}{3}$ m **(c)** 40 m **(d)** 20 m

3. From a point on level ground, 14 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $14\sqrt{3}$ m **(b)** 14 m **(c)** $\frac{14\sqrt{3}}{3}$ m **(d)** $28\sqrt{3}$ m

4. From a point on level ground, 24 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 8 m **(b)** $24\sqrt{3}$ m **(c)** $8\sqrt{3}$ m **(d)** 24 m

5. A vertical pole is 10 m high. The angle of elevation of the sun is 30° and the pole casts a shadow on level ground. The length of the shadow is: 1

(a) 10 m **(b)** $10\sqrt{3}$ m **(c)** 20 m **(d)** $\frac{10\sqrt{3}}{3}$ m

6. From the top of a cliff 15 m high, the angle of depression of a boat on the sea is 30°. The horizontal distance of the boat from the foot of the cliff is: 1

(a) $5\sqrt{3}$ m **(b)** $15\sqrt{3}$ m **(c)** 15 m **(d)** $30\sqrt{3}$ m

7. A ladder 18 m long leans against a vertical wall and makes an angle of 30° with the level ground. The height reached on the wall is: 1

(a) 9 m **(b)** $9\sqrt{3}$ m **(c)** 18 m **(d)** $6\sqrt{3}$ m

8. As the sun rises higher in the sky, the shadow of a vertical pole: 1

(a) stays the same **(b)** becomes shorter

(c) becomes zero at once **(d)** becomes longer

9. From a point on level ground, 25 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) $\frac{25\sqrt{3}}{3}$ m **(b)** 50 m **(c)** 25 m **(d)** $25\sqrt{3}$ m

10. From a point on level ground, 25 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $50\sqrt{3}$ m **(b)** 25 m **(c)** $\frac{25\sqrt{3}}{3}$ m **(d)** $25\sqrt{3}$ m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The string of a kite is 24 m long. It is taut and makes an angle of 60° with the level ground. Find the height of the kite above the ground. 2

12. From a point on level ground, 13 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. Find the height of the tower. 2

13. A vertical pole is 12 m high. The angle of elevation of the sun is 30° and the pole casts a shadow on level ground. Find the length of the shadow. 2

14. The string of a kite is 30 m long. It is taut and makes an angle of 60° with the level ground. Find the height of the kite above the ground. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 30 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find the height reached on the wall and the distance of the foot of the ladder from the wall (in this order). 3

16. From the top of a cliff $20\sqrt{3}$ m high, the angle of depression of a boat on the sea is 30°. Find the horizontal distance of the boat from the foot of the cliff. 3

17. A vertical pole is $8\sqrt{3}$ m high. The angle of elevation of the sun is 30° and the pole casts a shadow on level ground. Find the length of the shadow. 3

OR

A ladder 40 m long leans against a vertical wall and makes an angle of 30° with the level ground. Find the height reached on the wall and the distance of the foot of the ladder from the wall (in this order). 3

18. From the top of a cliff $9\sqrt{3}$ m high, the angle of depression of a boat on the sea is 30°. Find the horizontal distance of the boat from the foot of the cliff. 3

OR

A vertical pole is $11\sqrt{3}$ m high. The angle of elevation of the sun is 60° and the pole casts a shadow on level ground. Find the length of the shadow. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two points on level ground lie on opposite sides of a tower of height $18\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. Find the distance between the two points. 5

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 18 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower and the initial distance of the point from its foot (in this order). 5

20. Two points on level ground lie on opposite sides of a tower of height $21\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. Find the distance between the two points. 5

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 22 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower and the initial distance of the point from its foot (in this order). 5

END OF QUESTION PAPER — Total: 40 Marks